VERITAS UNIVERSITY, ABUJA

SEN307 — DATABASE SYSTEMS

WEEK 4 PRACTICAL: DATABASE DESIGN & NORMALIZATION

Lab 2 | Normalization from First Normal Form (1NF) to Third Normal Form (3NF)

Student details	Information
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Department	Software Engineering
Practical	Week 4 — Lab 2

1. Introduction

The purpose of this practical is to design a relational database for a university course-registration system involving students, courses, lecturers, departments and enrolments. The database is normalized to reduce repeated data and improve consistency. The final design uses primary keys (PKs) to identify records and foreign keys (FKs) to connect related tables.

2. Original Unnormalized Registration Table

The handout's registration sheet contains multi-valued cells for course codes, course titles, lecturers and lecturer departments. These repeating groups make the data difficult to query and maintain.

matric_no	student_name	student_dept	course_codes	course_titles	lecturers	lecturer_depts	session
20231234	Ada Obi	Software Engineering	SEN307, SEN301	Database Systems; Software Architecture	Dr A. Okafor; Dr M. Bello	Software Eng.; Software Eng.	2026/2027
20231235	John Musa	Computer Science	SEN307, COS301	Database Systems; Operating Systems	Dr A. Okafor; Mrs F. Eze	Software Eng.; Computer Sci.	2026/2027
20231236	Grace Okon	Software Engineering	SEN301	Software Architecture	Dr M. Bello	Software Eng.	2026/2027

3. First Normal Form (1NF)

To reach 1NF, each cell must contain a single value. Each student–course registration is therefore represented by its own row. The composite key is (matric_no, course_code, session), because no single column uniquely identifies every registration.

matric_no	student_name	student_dept	course_code	course_title	session
20231234	Ada Obi	Software Engineering	SEN307	Database Systems	2026/2027
20231234	Ada Obi	Software Engineering	SEN301	Software Architecture	2026/2027
20231235	John Musa	Computer Science	SEN307	Database Systems	2026/2027
20231235	John Musa	Computer Science	COS301	Operating Systems	2026/2027
20231236	Grace Okon	Software Engineering	SEN301	Software Architecture	2026/2027

What was fixed: Multi-valued cells were removed and every student–course pair was placed on a separate row. Student and course details are still repeated at this stage.

4. Second Normal Form (2NF)

The 1NF table has a composite key. Some non-key attributes depend on only part of that key: student_name and student_dept depend on matric_no, while course_title depends on course_code. These partial dependencies are removed by separating student, course and enrolment data.

Table	Attributes
Student	matric_no (PK), student_name, student_dept
Course	course_code (PK), course_title
Enrolment	matric_no (PK, FK), course_code (PK, FK), session (PK)

What was fixed: Student attributes and course attributes were moved to their own tables, leaving Enrolment to record which student takes which course in a particular session.

5. Third Normal Form (3NF)

To reach 3NF, department information is stored once in Department instead of being repeated in Student or Lecturer. Lecturer information is separated into Lecturer. CourseTeaching records which lecturer teaches which course in a given session, resolving the many-to-many relationship between lecturers and courses.

Table	Final attributes (PK and FK marked)
Department	dept_id (PK), dept_name
Student	student_id (PK), matric_no (UNIQUE), full_name, dept_id (FK → Department.dept_id)
Lecturer	lecturer_id (PK), lecturer_name, dept_id (FK → Department.dept_id)
Course	course_id (PK), course_code (UNIQUE), course_title
Enrolment	student_id (PK, FK → Student.student_id), course_id (PK, FK → Course.course_id), session (PK)
CourseTeaching	lecturer_id (PK, FK → Lecturer.lecturer_id), course_id (PK, FK → Course.course_id), session (PK)

What was fixed: Department details and lecturer details are stored separately, removing transitive dependencies and avoiding repeated department names. CourseTeaching stores lecturer–course assignments by session.

6. Relationships and Cardinalities

- Department to Student: one-to-many (1:M).
- Department to Lecturer: one-to-many (1:M).
- Student to Course: many-to-many (M:N), resolved by Enrolment.
- Lecturer to Course: many-to-many (M:N), resolved by CourseTeaching, with session recorded for each assignment.

7. Conclusion

The registration data has been reorganized from a table with repeating groups into related tables that follow the principles of 1NF, 2NF and 3NF. The design reduces duplicated data and uses primary and foreign keys to preserve relationships. In MySQL Workbench, the six tables and their relationships should be represented in the EER diagram and saved as lab2_model.mwb, then exported as lab2_erd.png.